

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (IT)

November 2012

IT207 Computer Networking-1

Date: 06.11.2012, Tuesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

## Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

## SECTION - I

- Q - 1 (a) What is Data Communication? Explain the component of data communication. [04]  
(b) List and explain the data communication standards in detail. [03]
- Q - 2 (a) What is Line Configuration? Differentiate between Point-to-Point and Multipoint configuration in network. [04]  
(b) What is topology? Explain the different types of topology in network. [06]  
(c) List and Explain the different transmission mode in network with example. [04]

## OR

- (c) List and Explain the different categories of network with example. [04]
- Q - 3 (a) Explain the ISO-OSI Layered architecture of network in detail. [08]  
(b) Explain Stop-and-Wait protocol of flow control in network. [04]

## OR

- (b) Explain the Line Discipline in data link control in brief. [04]  
(c) Define the term: ARQ. [02]

## OR

- (c) Compare Go-Back-n and Selective-Reject ARQ. [02]

## SECTION - II

- Q - 4 (a) List the different error detection methods and explain LRC with suitable example. [04]  
(b) What is Checksum? Explain in brief. [03]
- Q - 5 (a) Differentiate between Periodic and Aperiodic signals. [03]  
(b) List the types of digital-to-digital encoding techniques and explain anyone in detail. [06]

## OR

- (b) List the different types of digital-to-analog modulation techniques and explain anyone in detail. [06]

(c) Discuss guided transmission medium. [05]

OR

(c) Discuss unguided transmission medium. [05]

Q - 6 (a) Define the following term: [06]

1. Flow Control
2. Modulation
3. Wavelength
4. Reflection
5. Encryption
6. Routing

(b) Explain the cyclic redundancy check technique with suitable example. [04]

(c) Explain the Time-division multiplexing process in detail. [04]

OR

(c) Explain the Frequency-division multiplexing process in detail. [04]

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